

Terraform Smoke Test Plan

This plan deploys the AI Cloud Lab, verifies bootstrap and private access, then safely dismantles the test system.

0. Open the repository directory

Run Terraform from the directory containing `main.tf`, `variables.tf`, and the other configuration files:

```
Set-Location C:\GitHub\ai-cloud-lab
Get-ChildItem *.tf
```

If `Get-ChildItem *.tf` returns no files, stop and change to the correct repository directory before continuing.

1. Set the test variables

Set the AWS profile, region, and temporary Open WebUI password before running Terraform. The password is required and has no default.

```
$env:TF_VAR_aws_profile = "ai-cloud-lab"
$env:TF_VAR_aws_region = "us-east-1"
$env:TF_VAR_open_webui_admin_password = "YourTemporaryStrongPasswordHere"
```

You can also pass non-secret variables directly:

```
terraform plan `
  -var="aws_profile=ai-cloud-lab" `
  -var="aws_region=us-east-1"
```

2. Validate and preview

```
terraform init
terraform fmt -check -recursive
terraform validate
terraform plan
```

For an exact plan/apply pair:

```
terraform plan `
  -out ai-lab.tfplan `
  -var="aws_profile=ai-cloud-lab" `
  -var="aws_region=us-east-1"

terraform apply ai-lab.tfplan
```

The plan file can contain sensitive values. Do not commit it; remove it after use:

```
Remove-Item ai-lab.tfplan
```

3. Deploy

```
terraform apply `
  -var="aws_profile=ai-cloud-lab" `
  -var="aws_region=us-east-1"
```

Apply completes when EC2 is running, not necessarily when Ollama, the model, and Open WebUI are ready.

4. Set the unique SSM target

After apply, capture the instance ID in an environment variable:

```
$env:instance_id = terraform output -raw instance_id
```

Use that unique target for the SSM session:

```
aws ssm start-session `
  --target $env:instance_id `
  --region us-east-1 `
  --profile ai-cloud-lab
```

Inside the SSM shell, check bootstrap readiness:

```
if test -f /var/lib/ai-lab/ready; then
  echo READY
elif test -f /var/lib/ai-lab/failed; then
  echo FAILED
else
  echo NOT_READY
fi

tail -n 100 /var/log/ai-lab-bootstrap.log
sudo docker ps
ai-lab-status
```

Wait for **READY** before opening the browser tunnel. If the result is **FAILED**, inspect the bootstrap log before retrying.

The SSM shell is a Linux shell. Run Linux commands there; run PowerShell commands such as `curl.exe` from a separate Windows PowerShell window.

5. Open the private browser tunnel

Run this in a second PowerShell terminal:

```
aws ssm start-session `
  --target $env:instance_id `
  --document-name AWS-StartPortForwardingSession `
  --parameters portNumber="8080",localPortNumber="8080" `
  --region us-east-1 `
  --profile ai-cloud-lab
```

Open:

```
http://localhost:8080
```

Initial login credentials:

- Email: `admin@example.local` unless you changed `open_webui_admin_email` .
- Password: the value assigned to `$env:TF_VAR_open_webui_admin_password` before apply.
- Display name: `Lab Admin` unless you changed `open_webui_admin_name` .

Change the temporary password immediately after confirming access.

If local port 8080 is already in use, keep the remote port at 8080 and use a different local port:

```
aws ssm start-session `
  --target $env:instance_id `
  --document-name AWS-StartPortForwardingSession `
  --parameters "portNumber=8080,localPortNumber=8081" `
  --region us-east-1 `
  --profile ai-cloud-lab
```

Then open `http://localhost:8081` .

6. Stop or dismantle the test system

If you may test again later, stop the instance to avoid ongoing compute charges:

```
aws ec2 stop-instances `
  --instance-ids $env:instance_id `
  --region us-east-1 `
  --profile ai-cloud-lab
```

For permanent cleanup, review the destroy plan and then remove all Terraform-managed resources:

```
terraform plan -destroy `
  -var="aws_profile=ai-cloud-lab" `
  -var="aws_region=us-east-1"

terraform destroy `
  -var="aws_profile=ai-cloud-lab" `
  -var="aws_region=us-east-1"
```

After teardown, verify that the instance and security group are gone. Remove `terraform.tfvars` if it contains a real password, but keep `terraform.tfvars.example` .

Likely failure points

- AWS credentials are missing, expired, or lack EC2, IAM, security-group, or SSM permissions.
- The selected region has no default VPC.
- The account lacks `c7i.4xlarge` quota, or the selected subnet's AZ does not offer that instance type.
- The Session Manager plugin is not installed locally.
- Bootstrap is still downloading packages, Ollama, the model, or the Open WebUI image.
- Local port 8080 is already occupied; use local port 8081 for the tunnel.